

Chilawo Nchimunya Munene

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EDUCATION

Grambling State University

Jan 2024 – Dec 2027

Bachelor of Science in Computer Science, Biology Minor, GPA: 3.9/4.0

Grambling, LA

Relevant Coursework: Probability & Statistics, Linear Algebra, AI/Machine Learning, Deep Learning, Data Structures & Algorithms, Software Engineering

Honors & Awards: EICOP 2026 Finalist, Earl Lester Cole Honors Scholar, President's List, TechWise Fellow (Google/CMU), NASA L'SPACE Scholar, ColorStack Fellow

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, Java, JavaScript, C++

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, MLflow, Weights & Biases | Classification, CNNs, Transfer Learning, Feature Engineering, NLP, Computer Vision | LLMs & Agentic AI (Anthropic Claude API, prompt engineering, prompt caching, agent design)

Data Analytics: pandas, NumPy, exploratory data analysis, statistical modeling, data pipelines & preprocessing | PostgreSQL, Prisma, relational data modeling, REST API design | Tableau, PowerBI, Excel, Plotly, Streamlit

Tools: Git, Flask, React, Node.js, Vercel, Render | Claude Code, Cursor, GitHub Copilot

EXPERIENCE

Data Analytics Intern - Edwards Research & Innovation Center (ERIC)

Jun 2025 – Jul 2025 | Memphis, TN

- Engineered end-to-end ML pipelines in scikit-learn across 4 years and **3,000+ records**, automating feature engineering, preprocessing, and model training to reach **96% accuracy**
- Applied classification models to isolate the **4 strongest predictors** of student success; shipped interactive Plotly dashboards **adopted by City Leadership** to restructure Choose901 program design and their alumni data collection process
- Partnered with non-technical stakeholders to translate an open-ended question into concrete requirements, owning the work from problem framing through final stakeholder presentation

PROJECTS

Pona - Food Safety Analyzer (live, in active development) | Python, FastAPI, React, PostgreSQL, Tesseract

[Live](#) | [GitHub](#)

- Built and deployed a food safety analyzer that checks products against a user's sensitivity profile via barcode or label photo: **94.5% allergen recall** benchmarked across **22,799 allergen tags** on real product data
- Replaced a fine-tuned DistilBERT normalizer measuring **F1 = 0.00** with rule-based matching at 94.5%, then stripped torch, transformers, and faiss from the stack entirely
- Rebuilt OCR gating after finding the pipeline reported **93.7% confidence while dropping an allergen**, adding completeness checks and regression tests for every failure mode

AI Breast Cancer Detection | Python, PyTorch, scikit-learn, MLflow

[GitHub](#)

- Trained and benchmarked **5 model architectures** (LR, DT, SVM, KNN, RF) on the Wisconsin Diagnostic dataset, building the full pipeline from preprocessing through cross-validation, **95-99% accuracy** across all configurations
- Evaluated on F1, precision, and recall rather than accuracy alone, given the asymmetric cost of false negatives in diagnostic screening
- Tracked every experiment in MLflow for full reproducibility across configurations, making results auditable and reruns deterministic

Autonomous Research Summarizer Agent | Python, Anthropic Claude API, Flask, BeautifulSoup

[GitHub](#)

- Designed and shipped an **autonomous LLM agent** that polls a live marketplace, claims tasks, fetches and normalizes source documents, and returns structured 6-section output with no human intervention per job
- Specified the 6-section output schema around reader decision-making, and cut per-job API cost via prompt caching
- Engineered for reliability in production on Render: per-job error isolation so one failure never halts the loop, graceful conflict handling, and a health-check endpoint for uptime monitoring

LEADERSHIP EXPERIENCE

Senior Resident Assistant - Grambling State University

Aug 2024 – Present | Grambling, LA

- Manage **8 RAs and 200+ residents**; coordinate welfare escalations across **3+ campus departments** and lead community programming and conflict resolution for a diverse residential population